

Cross-Environment Adaptation for Wi-Fi CSI-Based Human Activity Recognition

B.Tech Final Year Project

ABSTRACT

Wi-Fi Channel State Information (CSI) describes how a wireless signal is distorted as it travels from a transmitter to a receiver. Human movement perturbs this signal in measurable ways, making it possible to recognise activities such as walking, standing and jumping using ordinary Wi-Fi hardware alone. Because the approach needs no camera and nothing worn on the body, and works in darkness, it suits privacy-sensitive applications such as elderly care and patient monitoring.

The obstacle to deployment is not raw accuracy but generalisation. CSI encodes the multipath structure of the particular room in which it was recorded, so the features a model learns are tied to that room's walls, furniture and device placement. A classifier that performs well where it was trained degrades substantially when moved elsewhere, and collecting a full labelled dataset for every room of installation is not economically realistic.

This project treats the required labelling effort as the quantity to be measured, asking not whether adaptation is possible but how much new-environment data it costs. Using the ReWiS public dataset, which contains CSI recordings of four activity classes collected in three distinct indoor environments, a compact convolutional neural network is first trained and evaluated within one environment to establish a baseline. It is then tested directly in the two unseen environments to quantify the accuracy lost to environmental change, and subsequently fine-tuned using a deliberately limited number of labelled samples per class from the new environment. As a control, models are also trained from scratch on the same limited samples, isolating the benefit contributed by pre-training.

The outcome is a characteristic curve relating recognition accuracy in a new environment to the volume of labelled data available from it, together with the sample count at which fine-tuning recovers a specified fraction of baseline performance. Every configuration is repeated over multiple random seeds and multiple random selections of the adaptation samples, with model selection performed strictly on validation data, so that results remain reliable in the low-data regime. The work delivers a reproducible experimental protocol and a practical estimate of the deployment cost of Wi-Fi CSI sensing in previously unseen environments.